

“We suffer more in
imagination than in reality.”

- Seneca

StoicQuotes.com



class starts @ 9:00pm

Agenda

- ① Snowflake vs BigQuery vs Amazon Redshift vs Databricks
- ② Introduction to streaming
- ③ Architecture of streaming - Lambda and Kappa Architecture
- ④ Introduction to Spark streaming
- ⑤ Traditional Dstream
- ⑥ Structured streaming

output modes, checkpoints, Kafka

Snowflake vs Databricks vs Amazon Redshift vs BigQuery

Core Architecture

AWS Redshift : A DWH provides coupled and decoupled solution

BigQuery : A serverless DWH

Snowflake : A cloud Data Platform. Primarily a DWH

Databricks : A cloud Data Platform. Primarily a advanced spark eng.

Supported Data Types

AWS Redshift : Structured and Semi-structured

BigQuery : Structured and Semi-structured

Snowflake : All types of data. Optimised for structured & semi

Databricks : All types of data. Optimised for semi & unstructured

Underlying Data Format

AWS Redshift : Proprietary columnar

BigQuery : Proprietary columnar

Snowflake : Proprietary columnar + iceberg tabular format

Databricks : Parquet + Delta lake table format

cloud Ecosystem

AWS Redshift : AWS

Big Query : GCP

Snowflake : Multi-cloud

Databricks : Multi-cloud

Primary use case [sweet spot]

AWS Redshift : Business Intelligence

Big Query : Adhoc and Interactive SQL Analytics

Snowflake : BI, Data sharing.

Databricks : AI and unified data

Streaming

Continuous flow of data

[Bytes, KB, MB]

Apache storm

AWS Kinesis

Apache Flink

GCP Dataflow

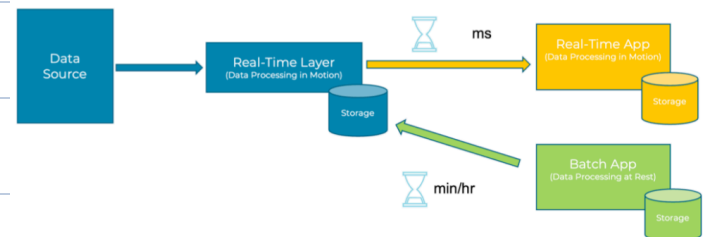
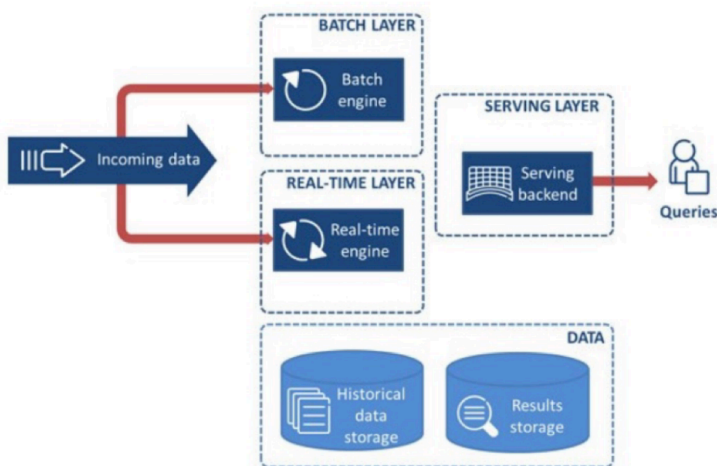
Apache Spark Streaming

Azure Stream Analytics

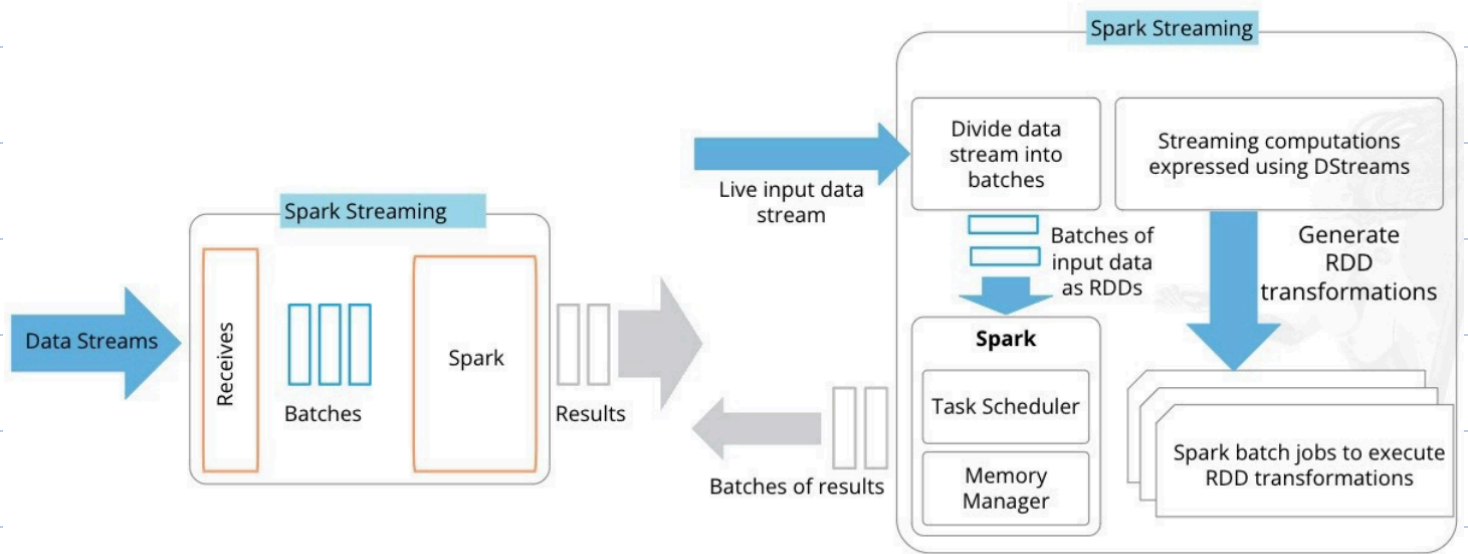
Apache Kafka stream

Streaming Architecture

Lambda Arch	Kappa Arch
① Both batch and Realtime Processing methods	① Only realtime processing to handle realtime & historical data
② 3 layers a Realtime (Speed layer) b Batch layer c Serving layer	② Single layer Architecture a Realtime (Speed layer) * possible through kafka
③ Storage for both methods	③ Storage at multiple level



Spark Streaming → near Realtime Data Processing



Stream Sources

- ① File Stream → text File Stream / binary Record Stream
- ② Socket Stream → socket Text Stream
- ③ Kafka / Flume / Kinesis → external dependency

Transformation

All RDD transformation → flatMap, map, filter, Reduce By Key

Output

Print to console

save As Text file

Netcat: A CLI for creating TCP connection to any machine

PySpark $\longrightarrow$ sql, stream, ml

Structured Streaming

Use High level API for performing any action or transformation on top of your stream input

from pyspark.sql.

entry point $\longrightarrow$ Spark session

Output Mode:

complete



table

2 more modes